x:NO:9 Implementation of Subnetting in CISCO packet Tracer.

Aim:

Implementation of subnetting in CISCO packet tracer simulator.

Creating a network topology:

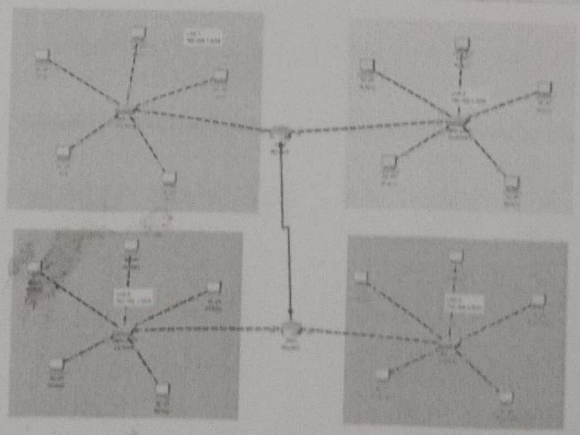
The first step in implementing classless IP subnetting is to create a network topology in packet tracer. To create a network topology in Packet tracer, select the "New" Button in the top left corner, then select "Network" and "generic".

Adding the Devices:

Once we have created our network topology, we can add devices to it. Here, we will be adding routers, switches and PC's. To add a device, select the device from the bottom left corner and drag it onto the network topology.

Subnetting:

To subnet the network address of 192.168.1.0/24 to provide enough space for at least 5 addresses for end devices, the switch and the router, we can use a /27 subnet mask.



The IP addressing for the network shown in the topology can be as follows:

Router R_1

Gigabit Ethernet 0/0: 192.168.1.1

Gigabit Ethernet 0/1: 192.168.2.1

Switch S_1:

Fast ethernet 0/1: 192.168.1.0/27

PC1: 192.168.1.11

PC2: 192.168.1.12

PC3: 192.168.1.13

PC4: 192.168.1.14

PC5: 192.168.1.15

Fast Ethernet 0/2: 192.168.2.0/27

PC1: 192.168.2.11

PC2: 192.168.2.12

PC3: 192.168.2.13

PC4: 192.168.2.14

PC5: 192.168.2.15

Router R_2:

Fast Ethernet 0/0: 192.168.3.1

Fast Ethernet 0/1: 192.168.4.1

Switch S_2:

Fast Ethernet 0/1: 192.168.3.0/27

PC1: 192.168.3.11

PC2: 192.168.3.12

PC3: 192.168.3.13

PC4: 192.168.3.14

PC5: 192.168.3.15

Configuring the devices:

Now that we have added our devices and connected them, we can start configuring them. We will start by configuring the router. Right-click on the router and select "CLI". This will open the command line interface (CLI) for the router.

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Testing the network:

Now that our network topology is configured, we can test the network; open a command prompt on each PC and try to ping the outer PC. If the ping is successful, then the network is functioning properly.

Student Observation:

a) Write down your understanding of subnetting.

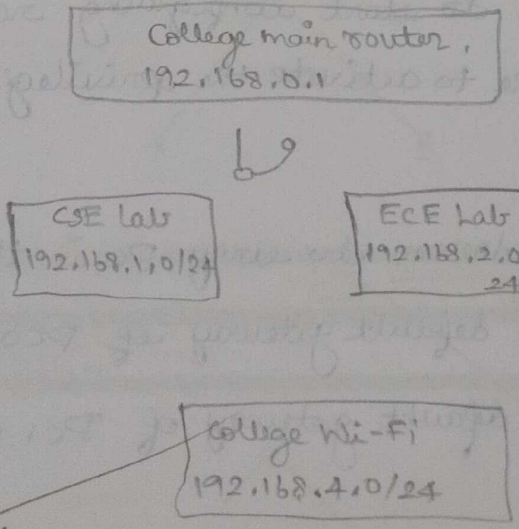
⇒ Subnetting is the process of dividing a large network into smaller logical networks called ~~network~~ subnets.

b) What are the advantages of implementing subnetting within a network?

⇒
* Reduces network traffic.
* Improves Security.
* Provides scalability.
* Simplifies network management.

c) Find out whether subnetting is implemented in your college. If yes, draw and list down the subnets used with IP addresses.

⇒ Yes, subnetting is commonly implemented in college networks to separate departments.



Conclusion:

Subnetting helps a college manage its network efficiently by dividing it into smaller networks.

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